

Regression Analysis

linear regression model

$$Y = \beta_0 + \beta_1 x + \epsilon$$

y is the predicted value of the dependent variable (y) for any given value of the independent variable (x).

B_0 is the intercept, the predicted value of y when the x is 0.

B_1 is the regression coefficient – how much we expect y to change as x increases.

x is the independent variable (the variable we expect is influencing y).

e is the error of the estimate, or how much variation there is in our estimate of the regression coefficient.

Definition

Given a collection of pairs (x, y) of numbers (in which not all the x -values are the same), there is a line $\hat{y} = \hat{\beta}_1 x + \hat{\beta}_0$ that best fits the data in the sense of minimizing the sum of the squared errors. It is called the least squares regression line. Its slope $\hat{\beta}_1$ and y -intercept $\hat{\beta}_0$ are computed using the formulas

$$\hat{\beta}_1 = \frac{SS_{xy}}{SS_{xx}} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where

$$SS_{xx} = \Sigma x^2 - \frac{1}{n}(\Sigma x)^2, \quad SS_{xy} = \Sigma xy - \frac{1}{n}(\Sigma x)(\Sigma y)$$

$\bar{x}$ is the mean of all the x -values, $\bar{y}$ is the mean of all the y -values, and n is the number of pairs in the data set.

The equation $\hat{y} = \hat{\beta}_1 x + \hat{\beta}_0$ specifying the least squares regression line is called the least squares regression equation.

Find the least squares regression line for the five-point data set

x	2	2	6	8	10
y	0	1	2	3	3

	x	y	x^2	xy
	2	0	4	0
	2	1	4	2
	6	2	36	12
	8	3	64	24
	10	3	100	30
Σ	28	9	208	68

$$SS_{xx} = \Sigma x^2 - \frac{1}{n}(\Sigma x)^2 = 208 - \frac{1}{5}(28)^2 = 51.2$$

$$SS_{xy} = \Sigma xy - \frac{1}{n}(\Sigma x)(\Sigma y) = 68 - \frac{1}{5}(28)(9) = 17.6$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{28}{5} = 5.6$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{9}{5} = 1.8$$

so that

$$\hat{\beta}_1 = \frac{SS_{xy}}{SS_{xx}} = \frac{17.6}{51.2} = 0.34375 \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 1.8 - (0.34375)(5.6) = -0.125$$

The least squares regression line for these data is

$$\hat{y} = 0.34375x - 0.125$$

THE ERRORS IN FITTING DATA WITH THE LEAST SQUARES REGRESSION LINE

$$e = y - \hat{y}$$

x	y	$\hat{y} = 0.34375x - 0.125$	$y - \hat{y}$
2	0	0.5625	-0.5625
2	1	0.5625	0.4375
6	2	1.9375	0.0625
8	3	2.6250	0.3750
10	3	3.3125	-0.3125

- 11-1.** An article in *Concrete Research* ("Near Surface Characteristics of Concrete: Intrinsic Permeability," Vol. 41, 1989), presented data on compressive strength x and intrinsic permeability y of various concrete mixes and cures. Summary quantities are $n = 14$, $\sum y_i = 572$, $\sum y_i^2 = 23,530$, $\sum x_i = 43$, $\sum x_i^2 = 157.42$, and $\sum x_i y_i = 1697.80$. Assume that the two variables are related according to the simple linear regression model.
- Calculate the least squares estimates of the slope and intercept.
 - Use the equation of the fitted line to predict what permeability would be observed when the compressive strength is $x = 4.3$.
 - Give a point estimate of the mean permeability when compressive strength is $x = 3.7$.
 - Suppose that the observed value of permeability at $x = 3.7$ is $y = 46.1$. Calculate the value of the corresponding residual.

$$11-1. \quad a) y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$S_{xx} = 157.42 - \frac{43^2}{14} \\ = 25.348571$$

$$S_{xy} = 1697.80 - \frac{43(572)}{14} \\ = -59.057143$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-59.057143}{25.348571} = -2.330$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{572}{14} - (-2.3298017)(\frac{43}{14}) = 48.013$$

$$b) \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$\hat{y} = 48.012962 - 2.3298017(4.3) = 37.99$$

$$c) \hat{y} = 48.012962 - 2.3298017(3.7) = 39.39$$

$$d) e = y - \hat{y} = 46.1 - 39.39 = 6.71$$

$$\text{Coefficient of correlation} = \frac{COV(XY)}{SD(X).SD(Y)}$$

$$\text{Where } Var(X) = \frac{\sum_{i=1}^n \left(x - \bar{x} \right)^2}{n-1}$$

$$Cov(XY) = \frac{\sum_{i=1}^n \left(x - \bar{x} \right) \left(y - \bar{y} \right)}{n}$$

Example

Consider the following observations on X and Y:

x 12 13 10 9 20 7 4 22 15 23

y 50 54 48 47 70 20 15 40 35 37

~~Est~~ Estimate the regression line, Compute coefficient of correlation, Standard error of estimates and

Coefficient of correlation

Solution

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
12	50	-1.5	8.4	-12.6	2.25	70.56
13	54	-0.5	12.4	-6.2	0.25	153.76
10	48	-3.5	6.4	-22.4	12.25	40.96
9	47	-4.5	5.4	-24.3	20.25	29.16
20	70	6.5	28.4	184.6	42.25	806.56
7	20	-6.5	-21.6	-140.4	42.25	466.56
4	15	-9.5	-26.6	-252.7	90.25	707.56
22	40	8.5	-1.6	-13.6	72.25	2.56
15	35	1.5	-6.6	-9.9	2.25	43.56
23	37	9.5	-4.6	-43.7	90.25	21.16
<u>135</u>	<u>416</u>	<u>0.0</u>	<u>0.0</u>	<u>445.0</u>	<u>374.5</u>	<u>2342.4</u>

$$\bar{x} = 135/10 = 13.5; \quad \bar{y} = 416/10 = 41.6$$

$$r = \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$b_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{445}{9}$$

$$n = 10 < 30$$

$$\text{Cov}(x, y) = 49.4444$$

$$\text{Var}(x) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = \frac{374.5}{9} = 41.6111$$

$$b_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = \frac{49.4444}{41.6111} = 1.1882$$

$$b_0 = \bar{y} - b_1 \bar{x} = 41.6 - (1.1882 \times 13.5) = 25.5593$$

Estimated regression line

$$\hat{y}_{ie} = b_0 + b_1 x_i$$

$$y_{ie} = b_0 + b_1 x_i$$

$$= 25.5593 + 1.1882 x_i$$

Coefficient of Correlation

$$r = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}$$

$$\text{SD}(X) = \sqrt{41.6111} = 6.4507 ; \text{SD}(Y) = \sqrt{\frac{\sum (y_i - \bar{y})^2}{N-1}} = \sqrt{\frac{2342.4}{9}}$$

$$\text{SD}(Y) = 16.1328$$

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$$r = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)} = \frac{49.4444}{(6.4507)(16.1328)} = 0.4751 < 1$$

positive relationship